IRON HACK

Data Analytics

Understanding & Boost Consumer Behavior for Online Shopping

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1. Introduction

Background and motivation

E-commerce continues to transform retail, making it essential to understand what drives user engagement on online platforms. Online shopping generates vast amounts of behavioral data, from clicks and page views to product interactions and navigation patterns. Analyzing this data helps businesses uncover how users explore products, what captures their attention, and which factors encourage deeper engagement or lead to abandonment. Understanding user behavior is critical for improving website design, optimizing product placement, tailoring marketing campaigns, and ultimately increasing conversions and customer satisfaction.

This project analyzes clickstream data from a European clothing website collected between April and August 2008, capturing 165,474 clicks across 24,026 user sessions. The dataset includes users from 47 countries interacting with 217 unique products in four main categories, providing insights into browsing behavior, product preferences, visual presentation, and seasonal patterns.

The motivation is threefold. First, identifying factors that influence engagement can help businesses optimize website design, product placement, and marketing strategies to increase conversions. Second, predictive models enable proactive personalization, allowing targeted promotions or recommendations for highly engaged users. Third, statistical analysis provides evidence-based guidance on product presentation, pricing, and seasonal campaigns, supporting data-driven decisions rather than intuition.

Research objectives

The project has three main objectives. First, exploratory analysis investigates user behavior across product categories, price ranges, and visual features using statistical tests and visualizations. Second, machine learning models classify sessions as high- or low-engagement, comparing algorithms like K-Nearest Neighbors, Random Forest, and Gradient Boosting, with performance optimized via hyperparameter tuning and cross-validation. Third, analytical findings are translated into actionable business recommendations on product placement, photography styles, pricing, and campaign timing, all grounded in statistical evidence and model insights.

The study addresses five key research questions:

(1) Does product price influence engagement?
(2) Do prominent page positions increase clicks?
(3) Does photography style affect user interactions?
(4) Which product categories drive the deepest browsing?
(5) Are there seasonal patterns in engagement from April to August that can inform marketing strategies?

2. Project Planning

Planning

I used a Trello dashboard, as shown in Figure 1, to organize and manage project tasks throughout the development process. It allowed me to break down large tasks into smaller checkpoints, track progress, and monitor deadlines in a clear and structured way. This approach proved useful, ensured transparency across project stages, and helped keep the project on schedule.

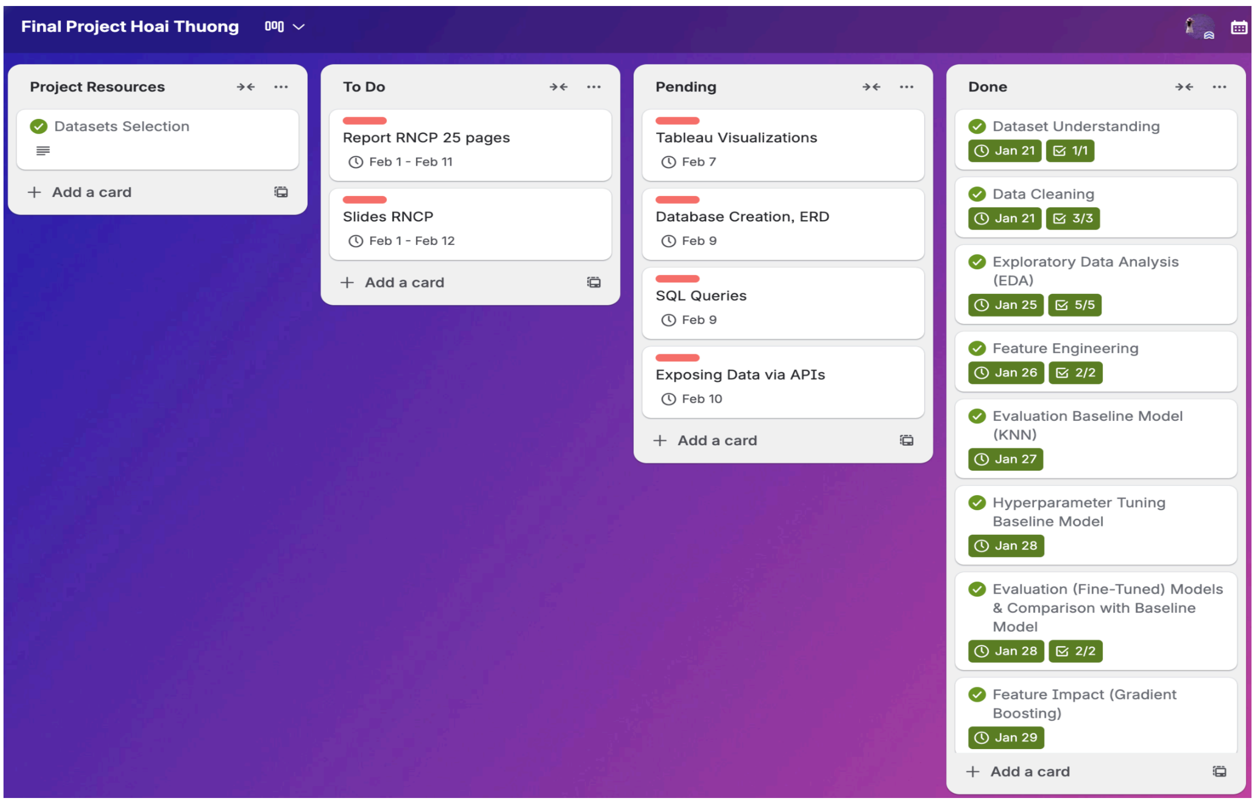


Figure 1: Trello dashboard used to manage daily project tasks.

Methodology

The project combined statistical analysis and machine learning to study user engagement. The development plan for this project was designed to ensure a systematic and effective analysis of user engagement on the e-commerce platform, encompassing the following key phases:

- **Data Collection**: Clickstream data from the European clothing website was aggregated, capturing 165,474 clicks across 24,026 sessions and 217 products. This dataset provides a foundation for analyzing browsing behavior, engagement patterns, and product preferences.

- **Data Cleaning & Exploratory Data Analysis (EDA)**: The raw dataset was cleaned and standardized to ensure completeness, consistency, and usability. Exploratory analysis, supported by visualizations such as box plots and bar charts, was conducted to identify patterns in user behavior, product interactions, and session characteristics. Statistical tests, including t-tests and ANOVA, were applied to support hypothesis-driven insights at a 5% significance level.
- **Database**: The processed data was structured for storage and efficient access, enabling organized management of click-level and session-level records for subsequent analysis.
- **Feature Engineering**: Click-level data was aggregated into session-level observations. Features capturing browsing depth, product diversity, visual interactions, navigation patterns, and categorical preferences were engineered to support predictive modeling of engagement.
- **Evaluation Model**: Machine learning models, including K-Nearest Neighbors, Random Forest, and Gradient Boosting, were developed for binary classification of high- versus low-engagement sessions. Data was split into an 80/20 stratified train-test set, and 5-fold cross-validation was used to ensure robust performance. Models were tuned using GridSearchCV and evaluated with accuracy, precision, recall, F1-Score, and ROC-AUC. Feature importance analysis provided insights into the drivers of user engagement.

All analysis was conducted in Python using pandas and numpy for data handling, matplotlib and seaborn for visualization, scipy.stats for statistical tests, and scikit-learn for machine learning and model evaluation.

3. Data Collection

The dataset[1] in CSV format contains 165,474 click events from 24,026 unique sessions and 217 clothing products, recorded between April and August 2008. Each row represents a single click, and clicks with the same session ID together form a complete browsing session, allowing us to analyze how users navigate the site, which categories they explore, and how deeply they engage. Figure 2 depicts the overview of the dataset.

[1] https://archive.ics.uci.edu/dataset/553/clickstream+data+for+online+shopping

```
df = pd.read_csv("../data/raw/e-shop_clothing_2008.csv", sep=";")
df
```

	year	month	day	order	country	session ID	page 1 (main category)	page 2 (clothing model)	colour	location	model photography	price	price 2	page
0	2008	4	1	1	29	1	1	A13	1	5	1	28	2	1
1	2008	4	1	2	29	1	1	A16	1	6	1	33	2	1
2	2008	4	1	3	29	1	2	B4	10	2	1	52	1	1
3	2008	4	1	4	29	1	2	B17	6	6	2	38	2	1
4	2008	4	1	5	29	1	2	B8	4	3	2	52	1	1
...	...	...	...	...	...	...	...	...	...	...	...	...	...	...
165469	2008	8	13	1	29	24024	2	B10	2	4	1	67	1	1
165470	2008	8	13	1	9	24025	1	A11	3	4	1	62	1	1
165471	2008	8	13	1	34	24026	1	A2	3	1	1	43	2	1
165472	2008	8	13	2	34	24026	3	C2	12	1	1	43	1	1
165473	2008	8	13	3	34	24026	2	B2	3	1	2	57	1	1

165474 rows × 14 columns

Figure 2: Overview of the dataset.

Overall, data quality was very good: no missing values, no duplicate rows, and consistent data types for dates, numbers, and IDs, as shown in Figure 3. However, some cleaning was still necessary to decode categorical variables, standardize column names, and ensure all values were in a clear, interpretable format for analysis.

Variable Name	Role	Type	Description	Units	Missing Values
year	Feature	Date	2008		no
month	Feature	Date	from April (4) to August (8)		no
day	Feature	Date	day number of the month		no
order	Feature	Integer	sequence of clicks during one session		no
country	Feature	Categorical	variable indicating the country of origin of the IP address		no
session ID	Feature	Integer	variable indicating session id (short record)		no
page 1 (main category)	Feature	Categorical	concerns the main product category		no
page 2 (clothing model)	Feature	Categorical	contains information about the code for each product (217 products)		no
colour	Feature	Categorical	colour of product		no
location	Feature	Categorical	photo location on the page, the screen has been divided into six parts		no

Figure 3: Detailed description of the dataset variables.

4. Data Cleaning & Exploratory Data Analysis (EDA)

Data Cleaning

The data cleaning process consisted of three main steps to prepare the raw dataset for analysis. The raw data were loaded from a CSV file into a DataFrame, importing all rows and columns while preserving their data types and maintaining the integrity of the original dataset.

First, all categorical variables were decoded using mapping information from the metadata file. Integer codes were converted into meaningful labels to improve interpretability. For example, country code *'29'* was mapped to *'Poland'*, category code *'1'* to *'trousers'*, color code *'2'* to *'black'*, and price level code *'2'* to *'below average'*. This decoding was applied consistently across all categorical columns.

Second, column names were standardized using a snake case naming convention to improve readability and consistency throughout the analysis pipeline. Columns containing spaces or special characters were renamed to concise and descriptive names, such as *'main_category'*, *'clothing_model'*, *'session_id'*, and *'price_usd'*.

Third, I performed data quality validation to ensure reliability. The dataset was checked for missing values and duplicate records, confirming that none were present. Data types were verified, descriptive statistics were reviewed to identify anomalies, and the sequential order of events within sessions was validated.

```
df.head()
```

day	order	country	session_id	main_category	clothing_model	colour	location	model_photography	price_usd	price_vs_category_avg	page
1	1	Poland	1	trousers	A13	beige	bottom in the middle	en face	28	below average	1
1	2	Poland	1	trousers	A16	beige	bottom right	en face	33	below average	1
1	3	Poland	1	skirts	B4	olive	top in the middle	en face	52	above average	1
1	4	Poland	1	skirts	B17	gray	bottom right	profile	38	below average	1
1	5	Poland	1	skirts	B8	brown	top right	profile	52	above average	1

Figure 4: Overview of the cleaned dataset.

The cleaned dataset, as shown in Figure 4, maintained its original size of 165,474 total clicks distributed across 24,026 unique sessions, with all 14 columns preserved but transformed into more interpretable formats.

Exploratory Data Analysis (EDA)

Hypothesis 1: Product price influences engagement

Our analysis shows that product price has a statistically significant impact on user engagement, measured by browsing depth. Sessions involving lower-priced products exhibited higher engagement, with an average of 7.03 clicks per session, compared to 6.75 clicks for sessions viewing higher-priced products. This represents a 4.2% increase in browsing depth for low-price sessions. The difference was confirmed through a Welch's t-test ($t = 2.44$, $p = 0.0146$), allowing us to reject the null hypothesis and conclude that lower product prices are associated with deeper browsing behavior. While the absolute difference may appear modest, it is meaningful when aggregated across tens of thousands of sessions. Visual analysis further showed that low-price sessions not only had a higher mean and median browsing depth, but also slightly greater variability, indicating that the effect is consistent across users and not driven by outliers.

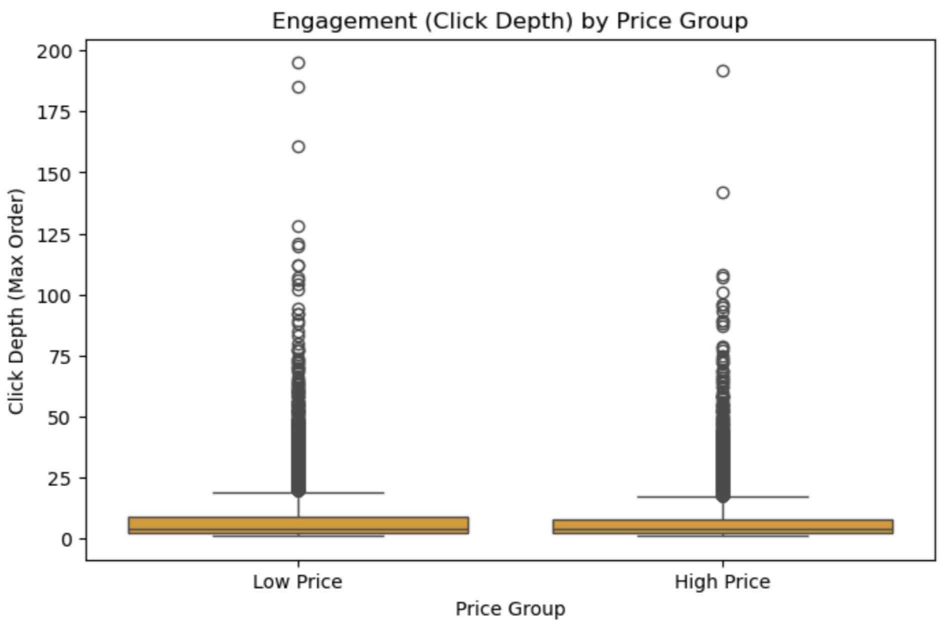


Figure 5: Engagement (click depth) across price groups.

> ***Insights**: Lower-priced products make users feel more comfortable browsing, so they explore more items and compare options. Showing budget-friendly products can increase user engagement and help attract users before recommending higher-priced items later.*

Hypothesis 2: Product presentation affects engagement

This analysis studied whether where a product appears on a webpage affects how much users interact with it. The page was divided into six positions (top and bottom rows, left to right). The results clearly show that products placed at the top of the page receive more attention. The top-left and top-middle positions together received over 40% of all clicks, even though they represent only a small part of the page. These positions also led to deeper browsing, with users clicking many more items after starting there. In contrast, bottom positions, especially bottom right, received far fewer clicks. This matches common browsing behavior, where users start reading from the top left and scan across and down the page. Overall, products placed in more visible areas are much more likely to be noticed and explored.

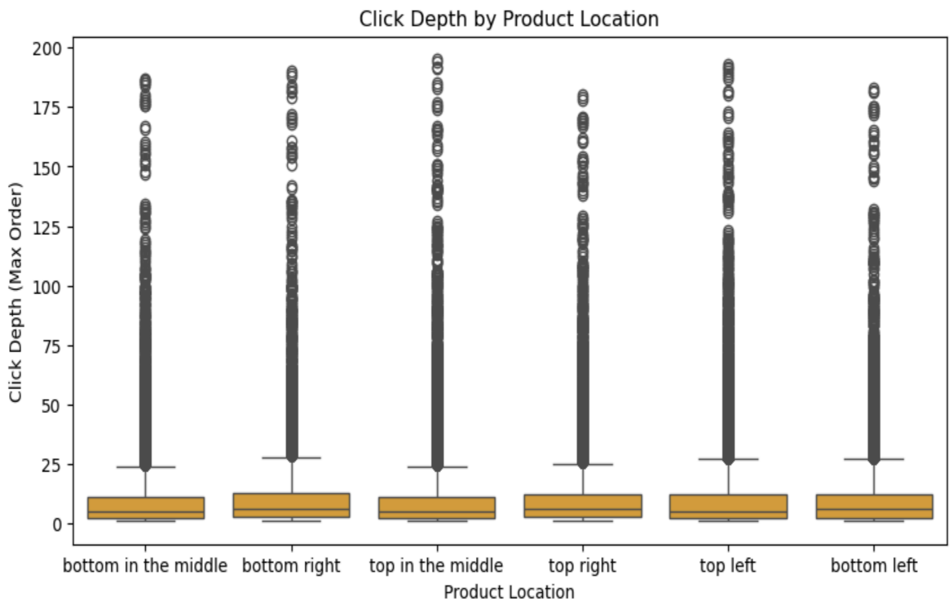


Figure 6: Click depth across product locations.

***Insights**: Users focus most on the top-left and top-middle areas of a webpage. Placing important products in these positions increases clicks and engagement, while products placed at the bottom, especially bottom right, are much less likely to be noticed.*

Hypothesis 3: Model photography influences early clicks

This analysis looked at whether product photo style affects when users click on a product during a browsing session. Two styles were compared: en face (front-facing) and profile (side view). The results show a clear difference. Products with en face photos are clicked earlier in the session than those with profile photos. On average, en face images were clicked almost 2 clicks earlier, meaning users notice and interact with them faster. This difference is statistically very strong and not due to chance. En face photos also made up most of the clicks in the dataset, suggesting they are more effective at grabbing attention. Profile photos, while clicked later, were linked to longer sessions, meaning users who click them tend to browse more carefully.

***Insights**: Front-facing product photos attract attention faster and lead to earlier clicks. They are ideal for catching users' interest quickly, while profile photos may be better for users who are browsing more carefully and closer to making a purchase.*

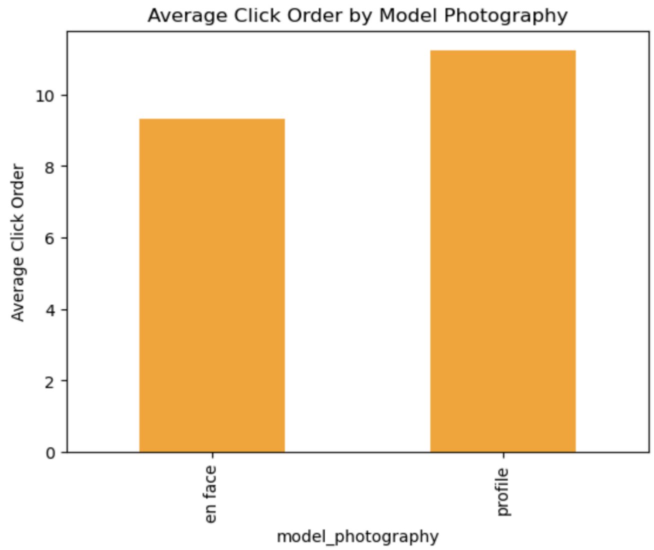


Figure 7: Average click order across model photography.

Hypothesis 4: Category drives browsing depth

This analysis studied how product category affects user engagement, measured by how many clicks users make in a session. Four categories were analyzed: sale items, blouses, trousers, and skirts. The results show strong differences between categories. Sale items generate the deepest browsing, with users clicking much more than in any other category. This suggests that users like to explore and compare many options when looking at discounted products. Blouses and trousers show medium engagement, with similar browsing depth, indicating a balanced shopping behavior. Skirts have the lowest browsing depth, meaning users tend to make faster decisions in this category. These differences are statistically significant and reflect distinct shopping behaviors across categories.

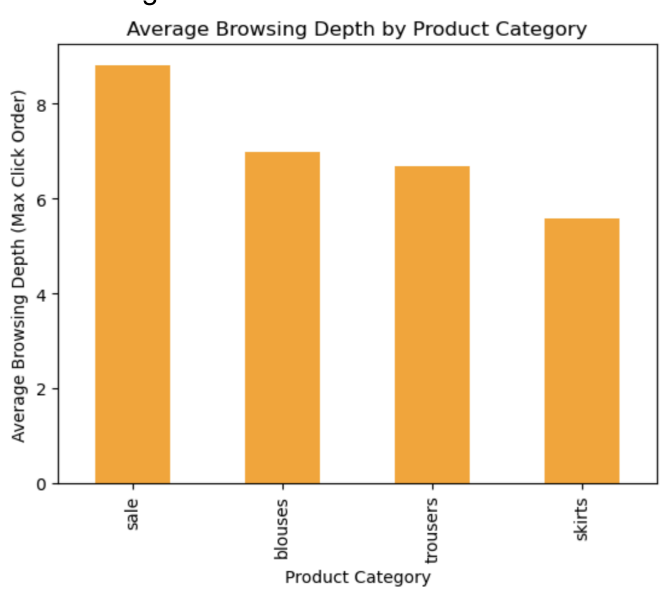


Figure 8: Average browsing depth across product categories.

***Insights**: Product category strongly influences browsing behavior. Sale items encourage deep exploration and comparison, while skirts lead to faster decisions. Sale sections can be used to drive engagement, while other categories can be optimized either for exploration (blouses, trousers) or quick conversion (skirts).*

Hypothesis 5: Seasonality affects browsing depth

This analysis looked at how user engagement changes by month, based on browsing depth. The data covers April to August. The results show a clear seasonal pattern. August has the highest engagement, with users browsing more deeply than in other months. June shows the lowest engagement, while April, May, and July remain relatively stable. These differences are statistically significant, meaning seasonality does affect browsing behavior. An interesting pattern also appears: although fewer users visit the site in summer, especially in August, those who do visit are more engaged and browse more products.

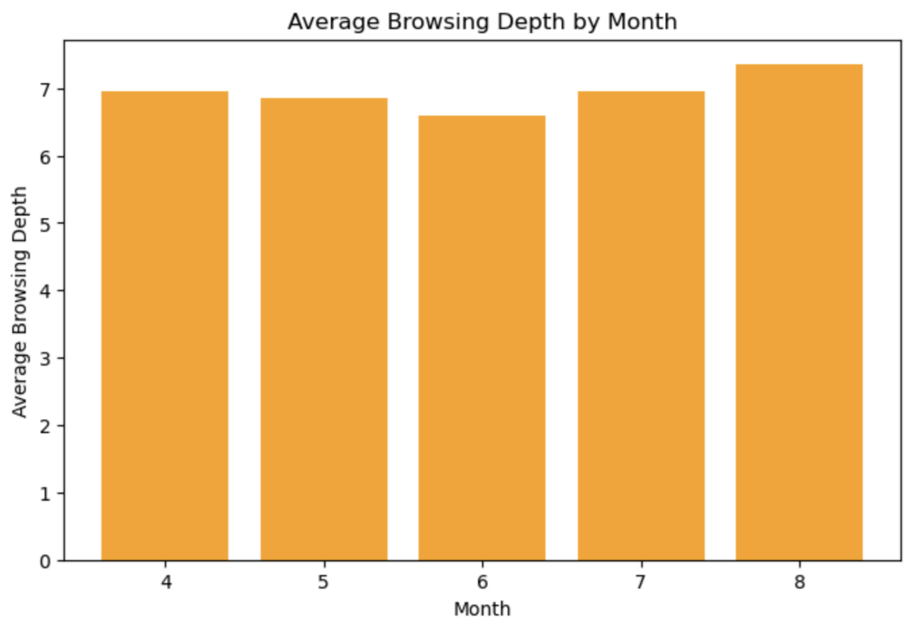


Figure 9: Average browsing depth across months.

***Insights**: User engagement varies by season. August attracts fewer users, but they browse more deeply, making it a strong period for promotions and product launches. June shows lower engagement and may need extra incentives or lighter marketing efforts.*

Tableau Visualizations

Figure 10 presents interactive Tableau visualizations used to explore, analyze, and communicate key insights from the cleaned dataset. The dashboards highlight trends, patterns, and performance metrics, enabling us to quickly understand the results and support data-driven decision-making based on the above hypotheses.

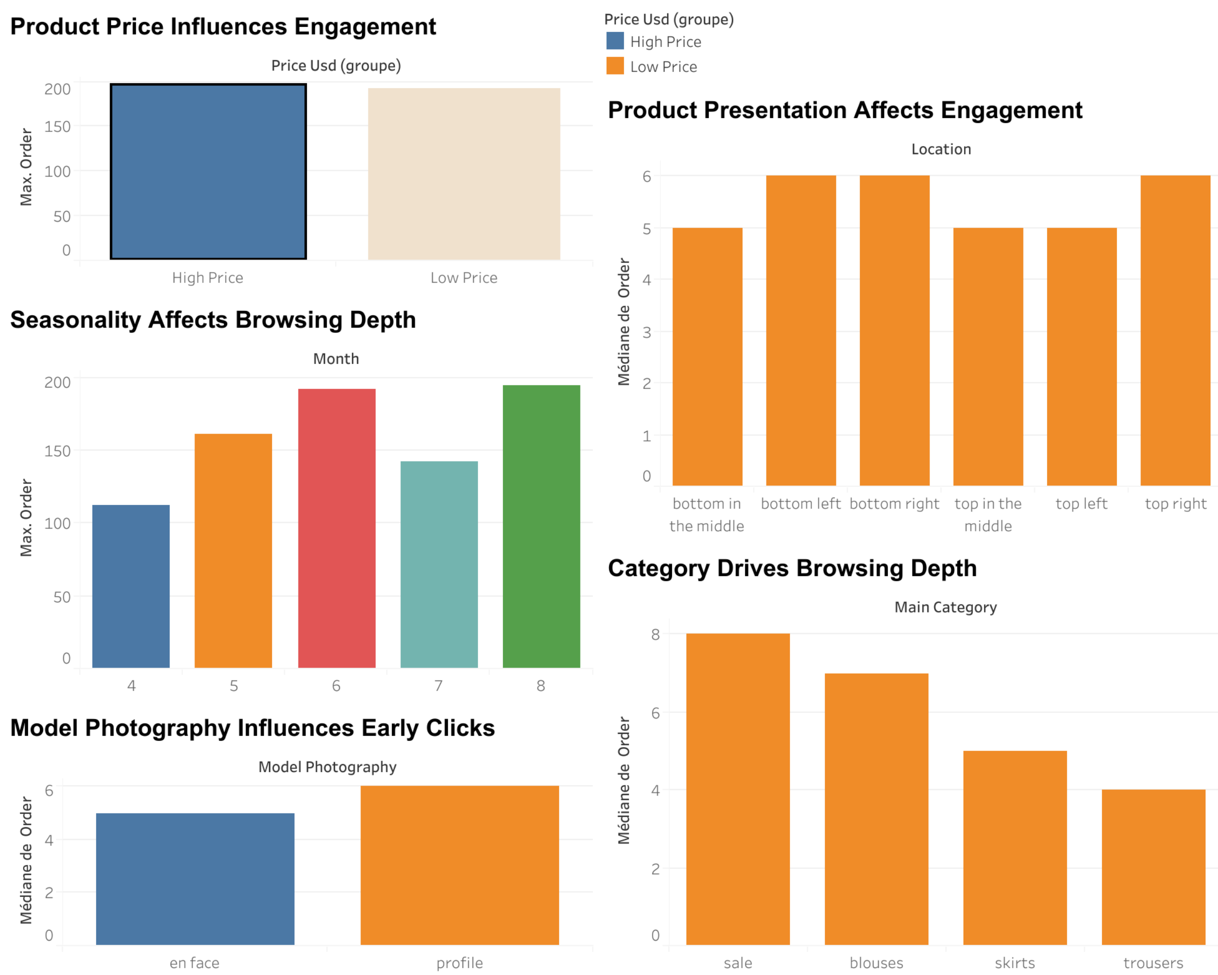


Figure 10: Tableau visualizations.

5. Database

Entity Relationship Diagram (ERD)

The Entity Relationship Diagram (ERD) illustrates the structure and relationships of the e-commerce dataset, as depicted in Figure 11 and Figure 12. It consists of three main tables: SESSIONS, CLICKS, and PRODUCTS.

- SESSIONS stores session-level information, including session ID, user country, and timestamp details.
- CLICKS records individual user interactions with products, linking each click to a specific session and product, along with click order, location, and photography style.
- PRODUCTS contains product-level details such as model, category, color, price, and average price comparison within the category.

The relationships are as follows: a single session can have multiple clicks (1-to-many), and each click is associated with one product (many-to-1). This ERD provides a clear overview of how user behavior, sessions, and product attributes are connected for analysis.

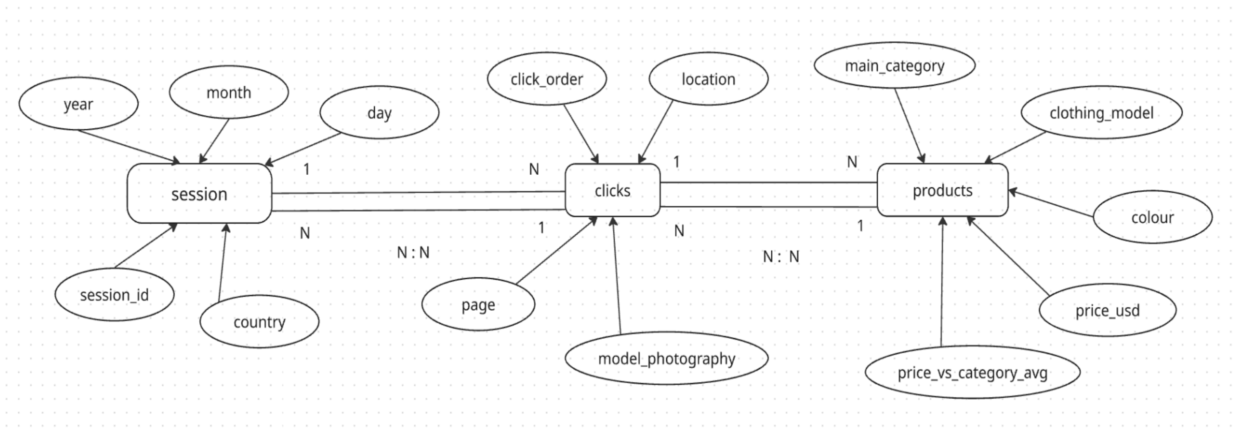


Figure 11: Entity Relationship Diagram (EDR).

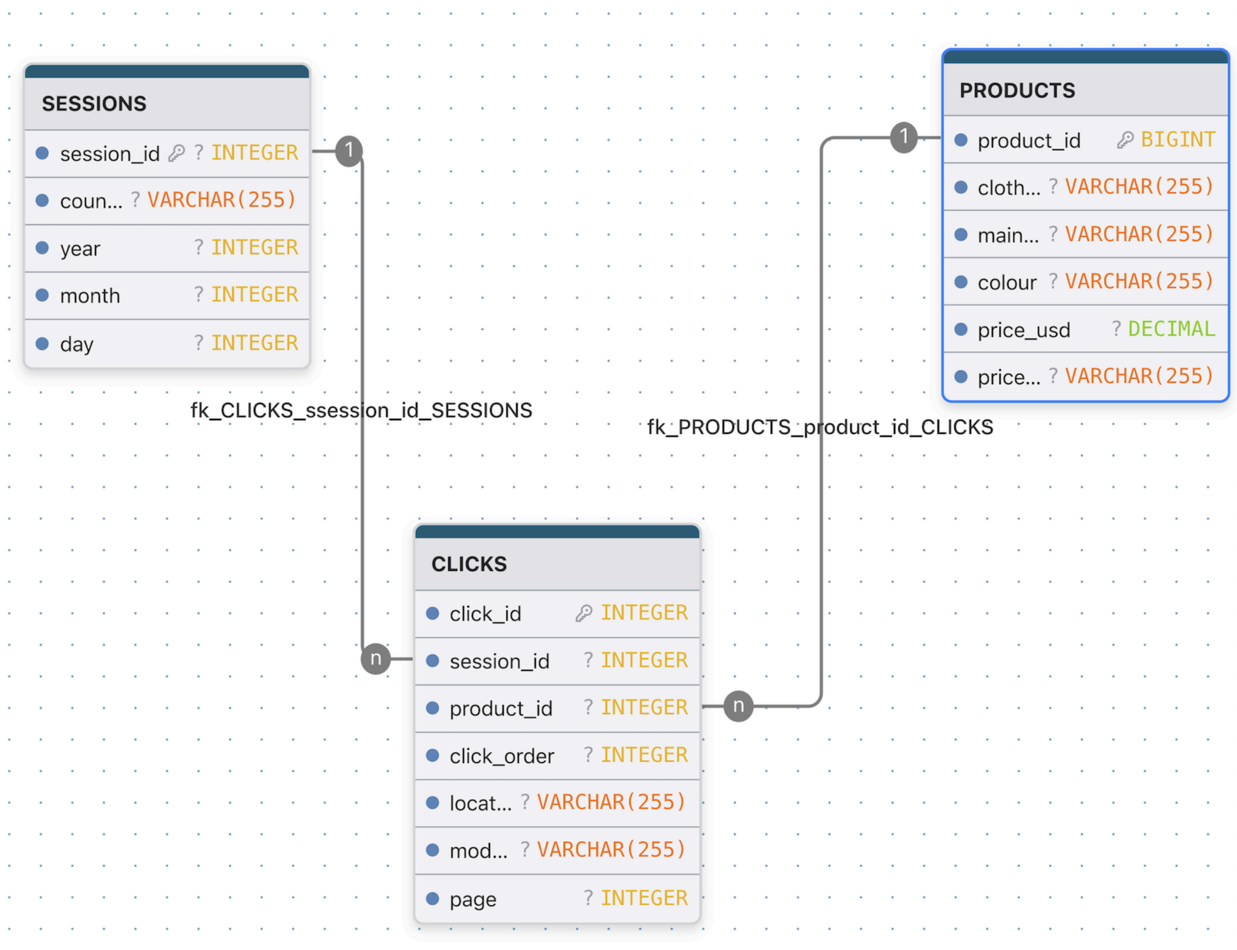
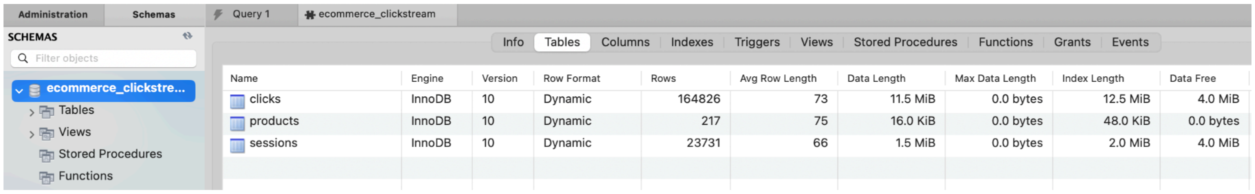


Figure 12: Database Entity Relationship Diagram.

Database Creation

The ‘ecommerce_clickstream’ database was created in MySQL Workbench to store three tables related to the e-commerce dataset.



Name	Engine	Version	Row Format	Rows	Avg Row Length	Data Length	Max Data Length	Index Length	Data Free
clicks	InnoDB	10	Dynamic	164826	73	11.5 MiB	0.0 bytes	12.5 MiB	4.0 MiB
products	InnoDB	10	Dynamic	217	75	16.0 KiB	0.0 bytes	48.0 KiB	0.0 bytes
sessions	InnoDB	10	Dynamic	23731	66	1.5 MiB	0.0 bytes	2.0 MiB	4.0 MiB

Figure 13: Database ecommerce_clickstream.

SQL Queries

To demonstrate how the database can be queried for analytical insights, I executed five SQL queries as follows.

Query 1: Top 10 most popular products by total clicks

```sql
-- Top 10 most popular products by total clicks
SELECT
    product_id,
    clothing_model,
    main_category,
    colour,
    price_usd,
    total_clicks,
    unique_sessions
FROM product_popularity
ORDER BY total_clicks DESC
LIMIT 10;
```

product_id	clothing_model	main_category	colour	price_usd	total_clicks	unique_sessions
3	B4	skirts	olive	52.00	3579	3212
54	A2	trousers	blue	43.00	3013	2557
30	A11	trousers	blue	62.00	2789	2454
15	P1	sale	blue	38.00	2681	2266
87	B10	skirts	black	67.00	2566	2415
84	A4	trousers	blue	38.00	2522	2224
91	A15	trousers	white	33.00	2489	2227
55	A5	trousers	blue	43.00	2354	1949
14	A10	trousers	blue	38.00	2280	1808
47	A1	trousers	navy blue	28.00	2265	2104

Figure 14: Results of Query 1.

Query 2: Sessions with highest engagement (most clicks)

```sql
-- Sessions with highest engagement (most clicks)
SELECT * FROM session_summary
ORDER BY total_clicks DESC
LIMIT 10;
```

product_id	clothing_model	main_category	colour	price_usd	total_clicks	unique_sessions
3	B4	skirts	olive	52.00	3579	3212
54	A2	trousers	blue	43.00	3013	2557
30	A11	trousers	blue	62.00	2789	2454
15	P1	sale	blue	38.00	2681	2266
87	B10	skirts	black	67.00	2566	2415
84	A4	trousers	blue	38.00	2522	2224
91	A15	trousers	white	33.00	2489	2227
55	A5	trousers	blue	43.00	2354	1949
14	A10	trousers	blue	38.00	2280	1808
47	A1	trousers	navy blue	28.00	2265	2104

Figure 15: Results of Query 2.

Query 3: Price analysis by category

```sql
-- Price analysis by category
SELECT
    main_category,
    COUNT(*) as product_count,
    MIN(price_usd) as min_price,
    AVG(price_usd) as avg_price,
    MAX(price_usd) as max_price
FROM products
GROUP BY main_category
ORDER BY avg_price DESC;
```

main_category	total_products	total_clicks	avg_clicks_per_product	total_sessions
trousers	43	49743	1156.8140	43305
sale	82	38747	472.5244	34996
blouses	59	38577	653.8475	35122
skirts	33	38407	1163.8485	34922

Figure 16: Results of Query 3.

Query 4: Click distribution by location

```sql
-- Click distribution by location
SELECT
    location,
    COUNT(*) as total_clicks,
    COUNT(DISTINCT session_id) as unique_sessions,
    ROUND(COUNT(*) * 100.0 / (SELECT COUNT(*) FROM clicks), 2) as percentage
FROM clicks
GROUP BY location
ORDER BY total_clicks DESC;
```

location	total_clicks	unique_sessions	percentage
top left	34532	13804	20.87
top in the middle	33383	14168	20.17
bottom in the middle	27783	12634	16.79
bottom left	27377	12436	16.54
top right	21656	10435	13.09
bottom right	20743	10221	12.54

Figure 17: Results of Query 4.

Query 5: Product color popularity

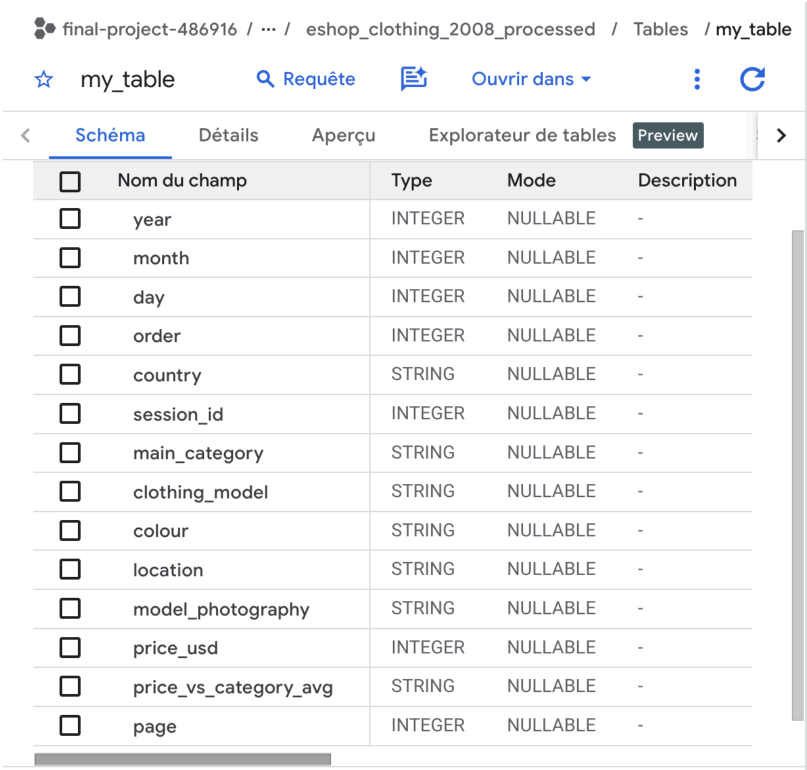
```sql
SELECT
    colour,
    COUNT(*) as product_count,
    SUM(total_clicks) as total_clicks,
    AVG(total_clicks) as avg_clicks_per_product
FROM product_popularity
GROUP BY colour
ORDER BY total_clicks DESC
LIMIT 10;
```

colour	product_count	total_clicks	avg_clicks_per_product
black	45	29764	661.4222
blue	18	29259	1625.5000
gray	27	17476	647.2593
brown	25	16517	660.6800
white	24	15939	664.1250
of many colors	22	13531	615.0455
red	14	8830	630.7143
beige	9	7785	865.0000
green	10	6876	687.6000
violet	9	6295	699.4444

Figure 18: Results of Query 5.

BigQuery

As a final method for data collection, a Big Data system was used in this project. Among the many available big data platforms, such as Snowflake, Amazon Redshift, and Oracle Database, Google BigQuery was selected due to its high scalability, fast query performance, ease of use, and cost effectiveness.



Nom du champ	Type	Mode	Description
year	INTEGER	NULLABLE	-
month	INTEGER	NULLABLE	-
day	INTEGER	NULLABLE	-
order	INTEGER	NULLABLE	-
country	STRING	NULLABLE	-
session_id	INTEGER	NULLABLE	-
main_category	STRING	NULLABLE	-
clothing_model	STRING	NULLABLE	-
colour	STRING	NULLABLE	-
location	STRING	NULLABLE	-
model_photography	STRING	NULLABLE	-
price_usd	INTEGER	NULLABLE	-
price_vs_category_avg	STRING	NULLABLE	-
page	INTEGER	NULLABLE	-

Figure 19: Table created in BigQuery.

A new table was created in BigQuery by loading the cleaned dataset, as shown in Figure 19. This allowed the data to be efficiently stored and queried using standard SQL. BigQuery also provides built-in support for data exploration and visualization, enabling the creation of different plots to support data analysis.

Figure 20 presents the results of a query analyzing country-level user behavior, which displays a list of countries with higher user engagement. This demonstrates how BigQuery can be used as a scalable analytical platform for processing and analyzing large datasets.

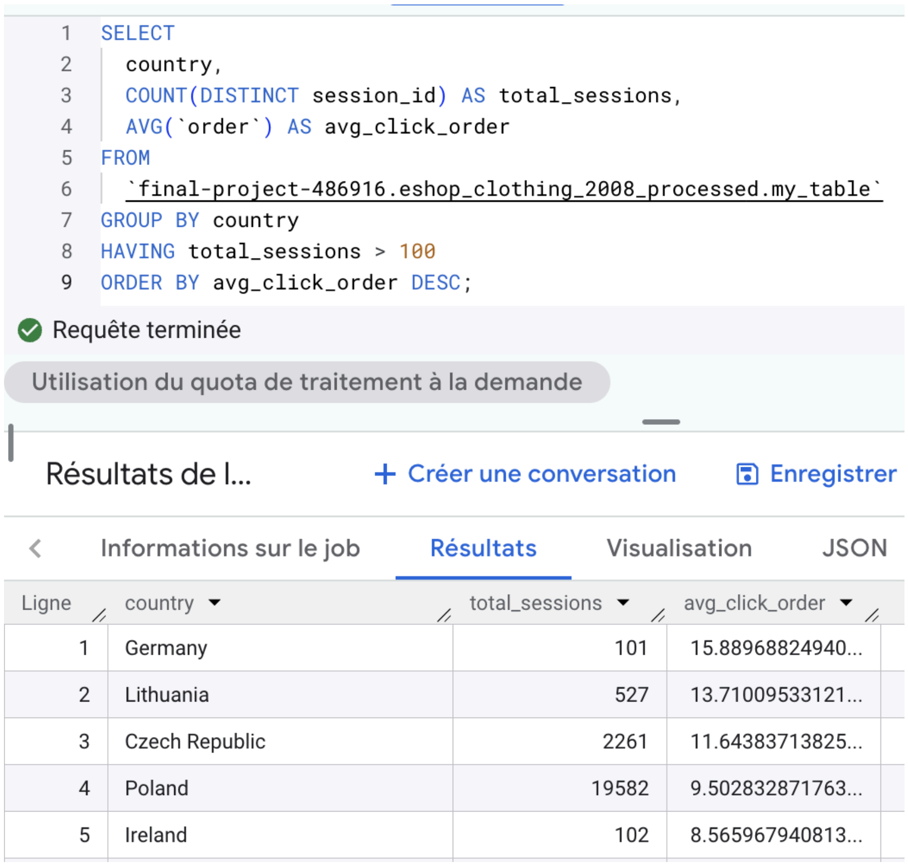


Figure 20: Results of country-level user behavior.

Exposing Data via APIs

To enable efficient access to the dataset stored in a MySQL database, a RESTful API was developed using Flask, a lightweight web framework written in Python. The API acts as an interface between the database and external users or applications, improving data accessibility and supporting analytical and research workflows. Flask was selected due to its simplicity, flexibility, and suitability for building RESTful services. The API follows a stateless client–server architecture, where each request contains all the information required for processing. Data is exposed through eight dedicated APIs, each designed to retrieve a specific portion of the dataset, including sessions, products and clicks. Table 1 shows a list of key RESTful APIs I implemented to access the database.

Table 1: List of RESTful APIs.

API	Description
GET api/sessions	Get all sessions
GET api/sessions/<id>	Get specific session details

GET /api/sessions/<id>/clicks	Get all clicks for a session
GET /api/products	Get all products
GET /api/products/<id>	Get specific product details
GET /api/products/popular	Get most popular products
GET /api/clicks	Get all clicks
GET /api/stats	Get database statistics

Figure 21 shows the results obtained from executing two GET API requests to retrieve all sessions in the dataset and the database statistics, respectively.

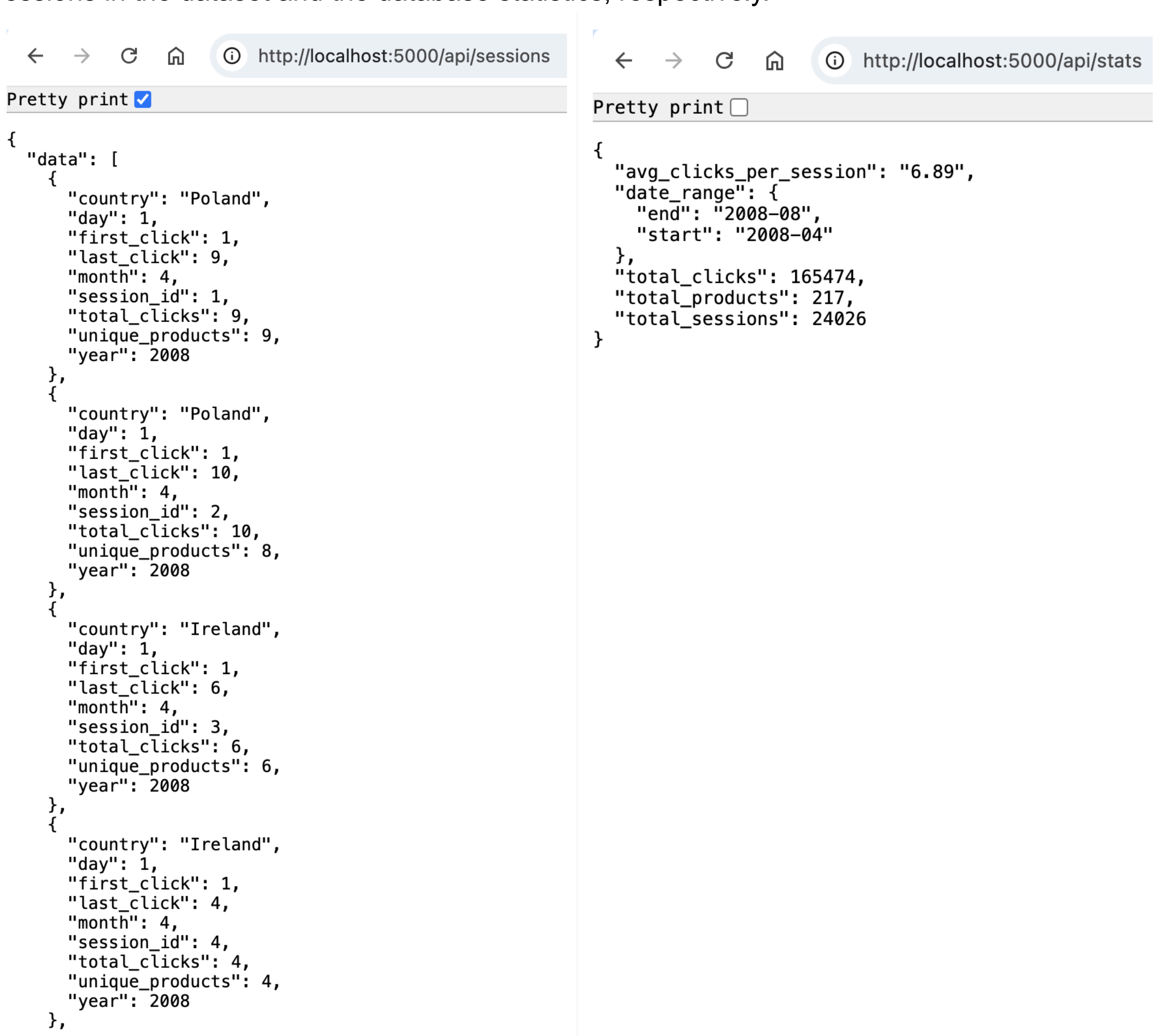


Figure 21: Results of APIs execution.

6. Feature Engineering

Session-based Datasets

For machine learning, the click-level dataset was aggregated to create a session-level dataset with 24,026 sessions, as the prediction target is session-level engagement. We engineered 31 features capturing different aspects of user behavior, divided into categorical and numerical types, as shown in Figure 22. Categorical features describe user preferences, such as country, initial and dominant product categories, color, product position, photography style, and first price level. Numerical features capture session behavior, including price statistics, category and color diversity, number of unique products and pages visited, navigation depth, temporal information, and counts of clicks by category, position, and style. These features aim to reflect both exploration breadth and engagement patterns, allowing the model to learn qualitative and quantitative relationships. Overall, this feature set is interpretable and provides rich information to predict which sessions are likely to have high engagement.

```
session_df.head()
```

	session_id	total_clicks	country	first_category	last_category	dominant_category	dominant_colour	first_location	dominant_location	photograpl
0	1	9	Poland	trousers	sale	skirts	gray	bottom in the middle	bottom in the middle	
1	2	10	Poland	skirts	sale	skirts	blue	bottom in the middle	bottom in the middle	
2	3	6	Ireland	skirts	sale	blouses	brown	bottom right	bottom right	
3	4	4	Ireland	trousers	blouses	blouses	black	bottom right	bottom right	
4	5	1	Czech Republic	blouses	blouses	blouses	white	top left	top left	

5 rows × 31 columns

Figure 22: Overview of the session-based dataset.

Label Creation

For the binary classification task, we defined sessions with 6 or more clicks as High Engagement and fewer than 6 clicks as Low Engagement. This threshold balances the classes (39% high, 61% low) while capturing meaningful user exploration, ensuring the model focuses on genuinely engaged sessions, as depicted in Figure 23. The dataset was split using a stratified 80/20 train-test split, maintaining the same class distribution in both sets to prevent evaluation bias. Stratification also ensures rare categories, such as users from less common countries, are represented proportionally, so the model sees a representative sample during training and testing. This labeling and splitting strategy provides a solid foundation for robust and interpretable machine learning.

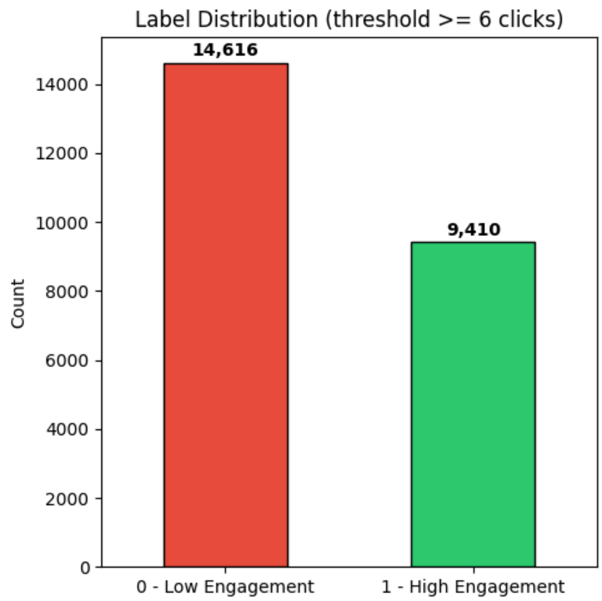


Figure 23: Label distribution of total clicks using a threshold of 6.

7. Evaluation Model

Baseline Model (K-Nearest Neighbors)

We established a K-Nearest Neighbors (KNN) classifier with default parameters as our baseline model to evaluate engagement prediction. KNN was chosen for its simplicity, interpretability, and speed, providing a reference point before testing more complex models. After preprocessing numerical features with scaling and categorical features with one-hot encoding, the model was trained on 19,220 sessions and tested on 4,806 sessions. As shown in Table 2, the baseline achieved 93.45% accuracy, 95.05% precision, 87.83% recall, 91.30% F1-Score, and 97.47% ROC-AUC, showing that the engineered features strongly predict session engagement. Cross-validation confirmed consistent performance with low variance, indicating good generalization. While precision was excellent, recall was slightly lower, meaning some high-engagement sessions were missed, but overall the baseline provides a robust reference and demonstrates that even a simple model can capture engagement patterns effectively.

Table 2: Baseline KNN model evaluation metrics.

Accuracy	0.934457
Precision	0.950546
Recall	0.878321
F1-Score	0.913007

ROC-AUC	0.974662

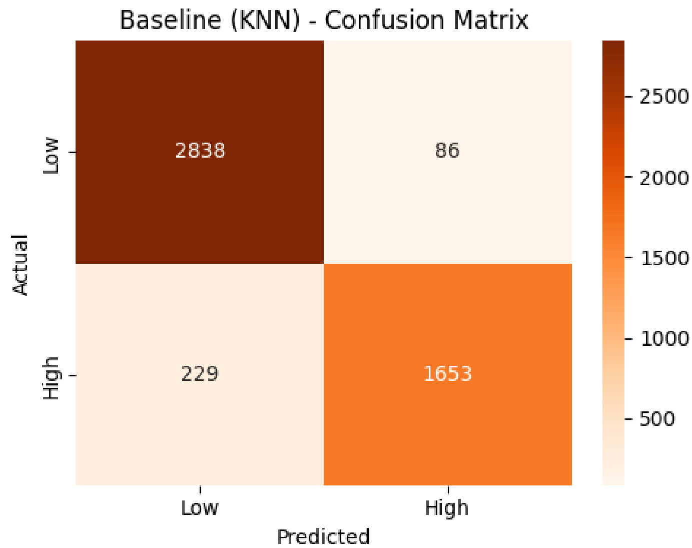


Figure 24: Confusion matrix of the baseline model KNN.

Hyperparameter Tuning (Cross Validation, Grid Search)

To improve model performance beyond the baseline, we applied systematic hyperparameter tuning using GridSearchCV with 3-fold stratified cross-validation, optimizing for F1-Score to balance precision and recall given the moderately imbalanced classes. For KNN, we tested different numbers of neighbors and weighting schemes, with the best configuration being 11 neighbors and uniform weighting. This modestly improved F1-Score to 91.99%, suggesting that KNN's performance is limited by its reliance on local similarity in feature space.

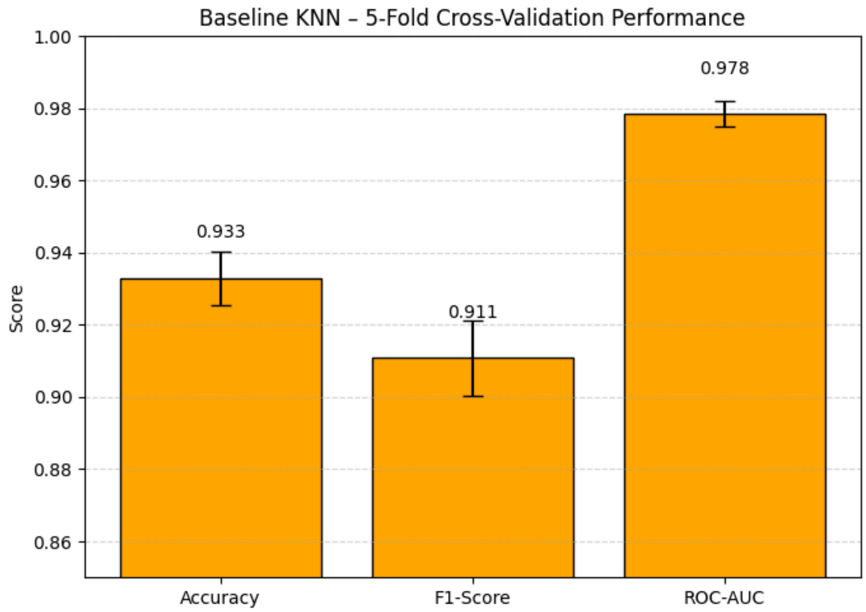


Figure 25: 5-fold cross validation performance of the baseline model KNN.

For Random Forest, we tuned the number of trees, tree depth, and splitting criteria. The optimal settings (200 trees, max depth of 20, min samples split 2, and min samples leaf 1) achieved an F1-Score of 98.66%, with near-perfect precision and recall, demonstrating the effectiveness of ensemble methods in capturing complex patterns.

Gradient Boosting tuning explored tree depth, learning rate, and number of boosting stages, with the best model using 200 estimators, learning rate 0.2, and depth 3. This model delivered the highest performance with an F1-Score of 99.39%, perfect precision, near-perfect recall, and flawless ROC-AUC, missing only 23 high-engagement sessions. Overall, hyperparameter tuning revealed that ensemble methods significantly outperform KNN, with Gradient Boosting emerging as the best model due to its sequential learning approach, ability to capture subtle patterns, and reliable performance for business applications.

Models Comparison

A comparison of the four models in Figure 26 shows clear performance differences. Gradient Boosting performed best, achieving 99.52% accuracy, perfect precision, 98.78% recall, an F1-Score of 99.39%, and a perfect ROC-AUC, representing an 8.85-point improvement over the baseline. Random Forest was a strong second, with 98.96% accuracy, 99.89% precision, 97.45% recall, and an F1-Score of 98.66%, confirming the effectiveness of ensemble methods. In contrast, KNN showed limited improvement, with 94.01% accuracy and 91.99% F1-Score, indicating difficulty in capturing complex user behavior patterns.

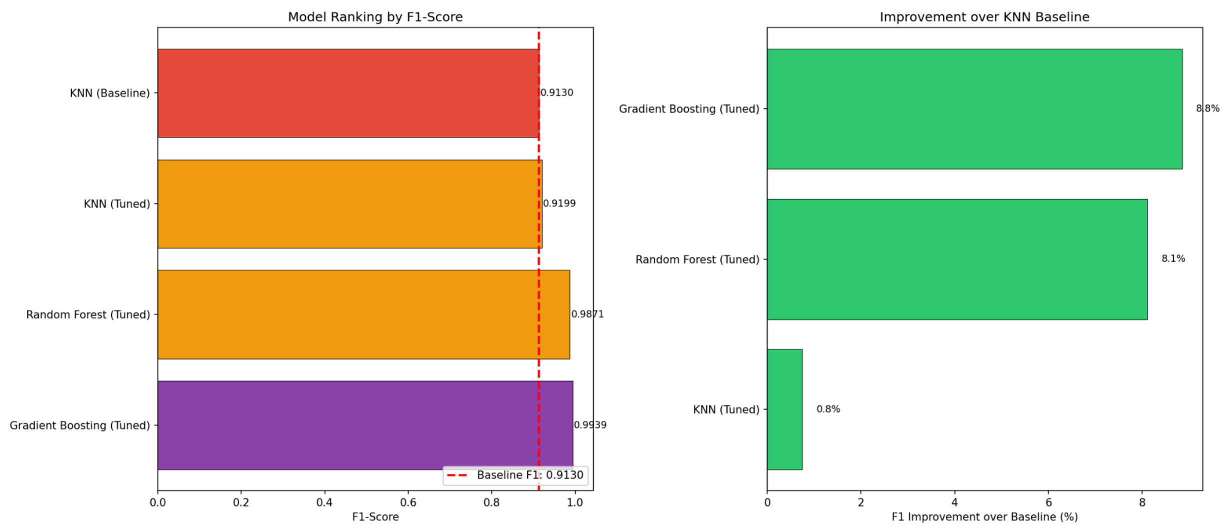


Figure 26: Performance comparison of all models against the baseline KNN model.

Overall, Gradient Boosting's perfect precision ensures no false positives, making it highly reliable for business applications such as targeted offers, while its near-perfect recall captures almost all high-engagement sessions. Random Forest delivers similar but slightly lower performance and may be preferred when simpler interpretation is required. KNN's weaker results suggest that distance-based methods are less suitable for this task. All models benefited from the engineered features, achieving F1-Scores above 90%, validating the feature design. Despite higher computational cost, Gradient Boosting was selected for deployment due to its superior performance and reliability.

Feature Impact

Analyzing feature importance from the Gradient Boosting model highlights which factors drive engagement predictions. As shown in Figure 27, the number of unique products viewed (*'unique_products'*) dominates with 92.7% importance, making it by far the strongest predictor of session engagement. This aligns intuitively, as sessions with more distinct product views naturally indicate deeper exploration. Other features, such as en_face_clicks (4.7%), top_position_clicks (0.6%), trouser_clicks (0.6%), and above_avg_price_clicks (0.4%), contribute only modestly, confirming that behavioral interactions matter more than static attributes like price or category.

Overall, the model shows that user behavior within a session is far more important than demographics, colors, or timing. Browsing diversity, captured by the number of unique products viewed, is the strongest predictor of engagement, while price and seasonal effects become negligible once behavior is considered. Other features, such as product position, photography style, and category, play a secondary but still meaningful role.

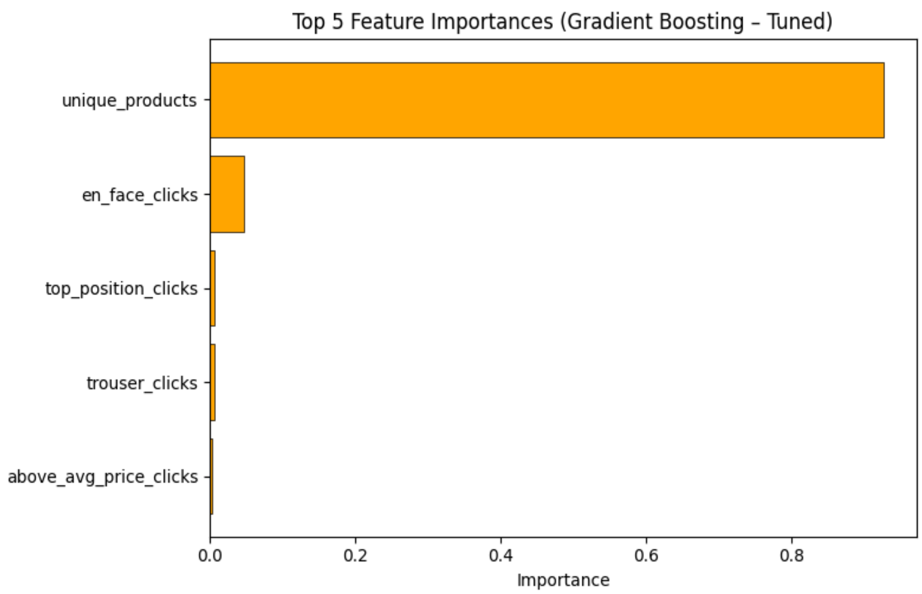


Figure 27: Top 5 feature importance of the tuned Gradient Boosting model.

From a business perspective, these findings suggest focusing on exposing users to a wider range of products rather than repeatedly showing similar items. Visual elements like frontal photography and top-page placement can help attract attention, but they are less impactful than encouraging exploration. The model can also be used in real time to detect high-engagement sessions and trigger personalized recommendations or discounts with high confidence, thanks to its perfect precision.

8. Conclusion

Key results

This project analyzed e-commerce clickstream data to understand user engagement and to build models that predict highly engaged browsing sessions. Both statistical analysis and machine learning produced clear and consistent results, offering strong support for data-driven decision-making in e-commerce.

The exploratory analysis confirmed all five research hypotheses. Lower-priced products generated slightly deeper browsing sessions, suggesting that affordable items encourage exploration. Product placement had a strong effect on engagement, with top-left and top-middle positions receiving a large share of clicks, confirming the importance of visual hierarchy on webpages. Photography style also mattered: en face images attracted clicks earlier in the session, while profile images were associated with deeper overall browsing. Product category showed large differences in engagement, with sale items producing much deeper sessions than other categories. Finally, engagement varied by season, with August showing higher browsing depth than earlier summer months.

The machine learning results were particularly strong. Among all models tested, Gradient Boosting performed best, achieving an F1-score of 99.39%, with perfect precision and near-perfect recall. This means the model almost never misclassified low-engagement sessions as high-engagement and successfully identified nearly all highly engaged users. Feature importance analysis showed that behavioral features dominated predictions, especially the number of unique products viewed, confirming that what users do during a session matters more than static attributes such as location or product metadata.

Overall, these results provide both strong predictive performance and clear behavioral insights that can be translated into practical business strategies, such as optimizing product placement, using effective photography styles, promoting sale items, and timing campaigns during high-engagement periods.

Challenges

Several challenges were encountered during the project:

- **Data understanding:** Many variables were encoded numerically and required external documentation to interpret correctly. Domain knowledge was needed to define meaningful engagement metrics.
- **Feature engineering:** Aggregating click-level data into session-level features required careful selection of aggregation methods to balance richness, interpretability, and model performance.
- **Label definition:** There was no predefined definition of "high engagement," so multiple thresholds were evaluated before selecting 6 clicks as a balanced and meaningful cutoff.
- **Generalization:** The data dates from 2008, so user behavior may differ from modern e-commerce contexts, especially due to mobile shopping.

Future work

Future work could extend this project in several directions. From a modeling perspective, removing or down-weighting dominant features such as '*unique_products*' could reveal more actionable insights, even if predictive accuracy decreases. Additional features such as time of day or day of week could capture further behavioral patterns. Advanced techniques such as model stacking or explainability methods could improve interpretability.

From a business perspective, deploying the model in real time would enable personalized interventions such as targeted promotions or recommendations during active sessions. A/B testing different engagement strategies on predicted high-engagement users could help convert engagement into purchases. Regular monitoring and retraining would ensure the model remains effective as user behavior evolves.

Finally, validating the findings on more recent data is essential. Future research could focus on predicting purchases rather than engagement alone, analyzing mobile versus desktop behavior, and segmenting users into behavioral personas. These extensions would further increase the practical value of the analysis.

GDPR

After a thorough assessment of the data used in this project, I can confidently confirm that no personal data was utilized at any stage. All data sources are entirely public and aggregated at the country level, ensuring full transparency and accessibility. Therefore, this project fully complies with the guidelines and principles of the General Data Protection Regulation (GDPR).

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- **Tableau Visualizations**: https://prod-ch-a.online.tableau.com/#/site/beaucielcosmetics-70b394878c/workbooks/416238?:origin=card_share_link